RESEARCH ARTICLE

Stroke Severity and Functional Benefit of Thrombectomy in Acute M2 Middle Cerebral Artery Occlusion

A Multicenter Cohort Study

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Abstract

Background and Objectives

Recent randomized trials reported no overall functional benefit of endovascular treatment (EVT) for distal medium-vessel occlusion (DMVO) and did not identify consistent effect modifiers to guide patient selection. Consequently, the role of EVT—particularly for M2 occlusions—remains controversial. We investigated whether baseline clinical severity modifies the association between successful recanalization and 90-day outcome in patients with M2 occlusion.

Methods

Multicenter retrospective cohort study at 2 tertiary stroke centers including consecutive adults with acute ischemic stroke due to M2 occlusion (January 2015–January 2023) triaged by multimodal CT and treated with EVT. The primary end point was functional independence (modified Rankin Scale [mRS] ≤ 2) at 90 days. Secondary end points included symptomatic intracerebral hemorrhage (sICH), mRS 0–1, and penumbra salvage volume (PSV). The primary analysis used multivariable logistic regression with baseline National Institutes of Health Stroke Scale (NIHSS) modeled linearly and an NIHSS×recanalization (modified Thrombolysis in Cerebral Infarction [mTICI] ≥2b) interaction term. Johnson-Neyman probing and inverse probability weighting (IPW) were applied; a supplementary restricted cubic spline analysis was performed to explore potential nonlinearity.

Results

Among 147 patients, 85 (58%) achieved successful recanalization. The mean age was 74 years (SD 13), and 47% were female. Higher baseline NIHSS (adjusted odds ratio [aOR] 0.83 per point, 95% CI 0.73–0.94) and older age (aOR 0.96 per year, 95% CI 0.94–0.99) were associated with lower odds of functional independence. NIHSS significantly modified the association between recanalization and outcome (interaction $p = 0.03$). The magnitude of the association between successful recanalization and functional independence was larger at higher NIHSS. Model-based estimates suggested a descriptive crossover around NIHSS 10, whereas statistical evidence of benefit emerged only at higher NIHSS values. Successful recanalization was associated with greater PSV (+33 mL, $p = 0.01$). In the PSV interaction model, onset-to-imaging time differed by recanalization status ($p < 0.01$): time was positively associated with PSV in the mTICI ≤2a group, but near zero in mTICI ≥2b. IPW analyses were concordant. sICH occurred in 8.2% vs 4.8% ($p = 0.42$).

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Glossary

aOR = adjusted odds ratio; **ATE** = average treatment effect; **CTA** = CT angiography; **CTP** = CT perfusion; **DMVO** = distal medium-vessel occlusion; **EVT** = endovascular treatment; **IPW** = inverse probability weighting; **IQR** = interquartile range; **mRS** = modified Rankin Scale; **NECT** = nonenhanced CT; **NIHSS** = National Institutes of Health Stroke Scale; **PSV** = penumbra salvage volume; **sICH** = symptomatic intracerebral hemorrhage.

Discussion

In M2 occlusions, the magnitude of the association between successful recanalization and functional independence was larger at higher NIHSS and not reliably demonstrable at lower NIHSS. These findings support a severity-informed, individualized EVT approach while remaining hypothesis-generating rather than prescriptive for specific NIHSS thresholds. Major limitations include the retrospective observational design and potential residual confounding.

Introduction

Acute distal/medium-vessel occlusion (DMVO) stroke represents ~25–40% of all ischemic strokes. Two recent randomized trials evaluated endovascular treatment (EVT) for DMVO and did not show overall clinical benefit of EVT over best medical treatment (BMT) when distal territories beyond the M1 segment were pooled.[1,2] In ESCAPE-MeVO, mortality was higher in the EVT arm (adjusted hazard ratio 1.82, 95% CI 1.06–3.12).

Recent DMVO trials enrolled many patients with mild deficits and did not identify robust severity-based effect modifiers. In DISTAL, the median National Institutes of Health Stroke Scale (NIHSS) was 6 and 41% of patients had NIHSS ≤5; in ESCAPE-MeVO, effect estimates increased numerically across NIHSS subgroups, but formal interaction testing was not significant. These data left uncertainty as to whether patients with more severe M2 deficits might still derive greater benefit from successful reperfusion.[3]

In the aftermath of the DMVO trials, a controversial debate has sparked, particularly regarding the benefit or harm of EVT for M2 occlusions, which constitute the majority of DMVO strokes.[4,5] The purpose of this study was to identify pretreatment variables—with a prespecified focus on NIHSS—that modify the association between successful recanalization (modified Thrombolysis in Cerebral Infarction [mTICI] ≥2b) and 90-day functional independence (modified Rankin Scale [mRS] 0–2) in M2 occlusions treated with EVT.

Methods

Patients

Acute ischemic stroke patients admitted between January 2015 and January 2023 at 2 high-volume tertiary stroke centers were consecutively screened (University Medical Center Hamburg-Eppendorf and Stanford University Medical Center). All consecutive patients meeting eligibility were included.

The a priori defined inclusion criteria for this study were: (1) acute ischemic stroke patients triaged by multimodal CT imaging on admission (nonenhanced CT [NECT], CT angiography [CTA] and CT perfusion [CTP]) at the tertiary care center providing EVT; (2) occlusion of the M2 segment of the middle cerebral artery; (3) EVT procedure performed; (4) absence of intracranial hemorrhage on baseline imaging; (5) absence of significant imaging artifacts; and (6) available follow-up CT within 24–48 hours after admission.

Baseline Clinical and Imaging Variables

Clinical variables included: age, sex, admission NIHSS (assessed by certified stroke neurologists or supervised neurology residents using the standard NIHSS; the admission score was used for analyses), IV thrombolysis (yes/no), onset/last-known-well to imaging time. All data were retrieved from clinical documentation and available medical records. Prestroke mRS and detailed vascular risk factors/comorbidity profiles were not captured in a harmonized manner across both centers for the full study period and were therefore not included in the primary adjusted analyses. Imaging variables included: ASPECTS, CT-perfusion–derived ischemic core (rCBF <30%), hypoperfusion (Tmax >6s), and penumbra (hypoperfusion minus core).

For classification, prespecified dichotomizations were applied where relevant (e.g., mRS 0–2 vs 3–6; mTICI ≥2b vs 0–2a; mTICI 2c–3 as "complete recanalization").

Study End Points and Outcome Measures

Primary outcome was functional independence at day 90 defined as mRS scores of 0–2. The mRS was ascertained in person during clinic follow-up visits or by structured telephone interview, conducted by a physician or trained and certified stroke study nurse using a standardized script. Secondary outcomes were (1) symptomatic intracerebral hemorrhage (sICH) defined by the SITS-MOST protocol as parenchymal hemorrhage type 2 on the 22-48 hours posttreatment follow-up scan, combined with either a neurologic deterioration of 4 points or more on the NIHSS from baseline, or from the lowest NIHSS value between baseline and after 24 hours, or leading to death,[6] (2) mRS 0–1 at day 90, and (3) penumbra salvage volume (PSV), which was defined as the difference between baseline penumbra volume and net infarct growth to the total ischemic lesion volume on follow-up imaging,[7] and early neurologic change (NIHSS at 24 hours minus baseline).

For the present analysis, sICH status was abstracted from institutional stroke registries and confirmed by targeted review of the 22–48 hours of follow-up CT scans and corresponding clinical records. The SITS-MOST classification was made by board-certified neuroradiologists in conjunction with vascular neurologists at each center, who were blinded to the final 90-day functional outcome; disagreements were resolved by consensus.

The mRS scores were evaluated by a physician or a trained and certified neurology study nurse. EVT was performed through a femoral artery approach with or without general anesthesia performed by board-certified interventional neuroradiologists.

Image Analysis

Image analysis of this study was performed blinded to clinical data. CTP images were automatically processed using RAPID with standard thresholds (iSchemaView, Menlo Park, CA). Baseline ischemic core volume was identified as tissue with a relative cerebral blood flow of <30% compared with the mean cerebral blood flow measured in regions of both hemispheres that do not have Tmax delays as the control. Total hypoperfusion volume was defined based on the brain tissue volume with prolonged Tmax >6s. Penumbra volume was defined as the difference of total hypoperfusion volume and core volume. Final modified Thrombolysis in Cerebral Infarction (mTICI) scores were determined by the operating neurointerventionalist and validated by a second attending neuroradiologist. Recanalization terminology was standardized: "successful recanalization" = mTICI ≥2b; "complete recanalization" = mTICI 2c–3. The Alberta Stroke Program Early CT Score (ASPECTS)[8] was independently determined on each pretreatment admission NECT examination by one neuroradiologist and validated by another experienced attending neuroradiologist. Follow-up CT imaging following EVT was used to calculate the total infarct volume and consequently PSV.

Statistical Analysis

Metric variables were summarized using the median and interquartile range (IQR), and differences between groups were tested using the Mann–Whitney *U* test. Normality was assessed using the Shapiro–Wilk test. Categorical variables were presented as counts and percentages and compared using the χ^2 test. Patients with functional independence at 90 days (mRS 0–2) were compared with those with poor outcome (mRS 3–6).

Center was included as a prespecified fixed effect in all adjusted and weighted analyses. The primary analysis estimated the association between successful recanalization (mTICI ≥2b) and mRS 0–2 using multivariable logistic regression with robust (sandwich) standard errors. Baseline NIHSS was modeled as a continuous linear term, and an NIHSS×recanalization interaction term was included to assess effect modification. The reference cell was mTICI ≤2a; thus, the main NIHSS term pertains to the nonrecanalized stratum and the NIHSS × (mTICI ≥2b) term represents the incremental change in the NIHSS effect when recanalization succeeds. Prespecified adjustment covariates (all pretreatment) were age, sex, ASPECTS, intravenous thrombolysis, onset/last-known-well–to-imaging time, baseline ischemic core volume, and center.

For clinical interpretation, we derived NIHSS-specific predicted probabilities, odds ratios, and absolute risk differences from the fitted interaction model at representative NIHSS values (5, 8, 10, 12, 15, and 18), with 95% confidence intervals obtained by postestimation on the logit/log-odds scale and transformed as appropriate. We used Johnson–Neyman probing to identify the observed NIHSS range in which the lower 95% confidence limit for the recanalization odds ratio exceeded 1.0. As robustness analyses, we performed inverse probability weighting using prespecified pretreatment covariates and a supplementary restricted cubic spline interaction model. Further technical details on weighting diagnostics, truncation, augmented IPW, bootstrap procedures, and banded margins analyses are provided in the Supplement (eAppendix 1). To reduce measured confounding when estimating the average treatment effect (ATE) of achieving mTICI ≥2b within attempted EVT, we performed inverse probability weighting (IPW). Treatment assignment A = 1 (mTICI ≥2b) vs A = 0 (mTICI ≤2a) was modeled with logistic regression using pretreatment covariates (age, sex, NIHSS [splines], ASPECTS, intravenous thrombolysis, onset-to-imaging time, baseline ischemic core and hypoperfusion volumes, and center). Stabilized weights were constructed as the marginal probability of the observed treatment divided by the estimated conditional probability from the propensity model. We truncated weights at the 1st/99th percentiles (and at the 5th/95th in sensitivity analyses) and fit weighted logistic models for mRS 0–2 including the A×NIHSS interaction. We present balance diagnostics (standardized mean differences before and after weighting; target <0.10), propensity overlap plots, and the effective sample size after weighting. As a robustness check, we also report augmented IPW (doubly robust) models that add the same pretreatment covariates to the weighted outcome regression. IPW estimates were interpreted as associational and subject to the usual assumptions of no unmeasured confounding, positivity, and correct model specification.

As a sensitivity analysis of functional-form assumptions, we additionally modeled baseline NIHSS flexibly using restricted cubic splines and fit analogous interaction models; corresponding model-based predicted probabilities and odds ratios are reported in the Supplement (eAppendix 1).

In addition to pointwise margins, we performed a banded margins analysis to summarize the effect of recanalization in the NIHSS 10–12 range, where the model-based curves cross. We computed the band-averaged OR and ARD for mRS 0–2 by averaging model-based predicted probabilities across integer NIHSS values 10, 11, and 12, with 95% CIs obtained via the delta method. This analysis is descriptive and complementary to the J–N region of significance and is not used to define a treatment threshold.

We limited inference to the empirical NIHSS support and considered 2-sided $p < 0.05$ statistically significant.

All analyses were performed using Stata version 18.5 (StataMP, StataCorp LLC, College Station, TX).

Standard Protocol Approvals, Registrations, and Patient Consents

The study was approved by the local institutional review boards (Ethikkommission der Ärztekammer Hamburg, WF04/13; and Stanford University Administrative Panel on Human Subjects in Medical Research, eProtocol #37209), both of which waived informed consent for this retrospective analysis of anonymized routinely collected clinical data with no impact on patient management. Data protection complied with the EU General Data Protection Regulation and applicable German federal/state regulations at German sites and with HIPAA at US sites; HIPAA language is not applied to non-US centers. All procedures adhered to the Declaration of Helsinki.

Data Availability

All data underlying the present analyses are available from the corresponding author upon reasonable request.

Results

Study Cohort

Seven hundred twenty-eight ischemic stroke patients with anterior circulation vessel occlusion were screened of which 147 patients showed an acute occlusion of the M2 segment of the MCA and were analyzed. The median age and NIHSS were 77 (IQR: 63–83) and 11 (IQR: 6–17), and the median time from onset to imaging was 98 minutes (IQR: 70–155 min). Sixty-nine patients (47%) were female. The median ASPECTS was 8 (IQR: 7–10). The mean core and hypoperfusion volumes were 17 mL (SD: 30) and 90 mL (SD: 90). Eighty-four patients (57%) received treatment with intravenous alteplase. In 85 patients (58%), successful recanalization (mTICI 2b-3) could be achieved after a median number of retrievals of 1 (IQR: 1–2). After 24 hours, the median NIHSS difference from baseline was −4 (IQR: −8 to 0), and 72 patients (49%) exhibited functional independence at day 90. Secondary sICH occurred in 10 patients (6.8%). Patient characteristics are presented in Table 1.

Key pretreatment variables stratified by recanalization status are presented in Table 1. Patients with successful recanalization (mTICI ≥2b) had lower baseline NIHSS (median 10 vs 14, $p =$ 0.04) and were more likely to have received intravenous thrombolysis (65% vs 47%, $p = 0.03$), whereas age and ASPECTS did not differ significantly between groups (Table 1). Primary and secondary outcomes by recanalization status are presented in Table 1: mRS 0–2 at day 90 was achieved in 57% of the mTICI ≥2b group vs 39% of the mTICI ≤2a group ($p =$ 0.03), with no significant difference in sICH (8.2% vs 4.8%, $p = 0.42$).

Interaction Analysis

In the multivariable logistic regression with linear NIHSS and an NIHSS×recanalization term, higher baseline NIHSS and older age were independently associated with lower odds of functional independence. In the reference stratum without successful recanalization (mTICI ≤2a), each 1-point increase in NIHSS was associated with a 17% reduction in the odds of mRS 0–2 (adjusted OR 0.83, 95% CI 0.73–0.94, $p < 0.01$), and each additional year of age with a 4% reduction (adjusted OR 0.96, 95% CI 0.94–0.99, $p = 0.01$) (Table 2). The NIHSS×recanalization interaction term was significant (adjusted odds ratio [aOR] 1.19, 95% CI 1.01–1.39, $p = 0.03$), indicating that the adverse NIHSS slope was attenuated among patients with successful recanalization (mTICI ≥2b), with the overall NIHSS-outcome association in this group approaching null. This pattern is consistent with successful reperfusion mitigating much of the detrimental impact of higher baseline stroke severity. Exploratory spline-based stratum-specific local NIHSS slopes are presented in eTable 1. Univariable and multivariable regression analyses with and without interaction term for secondary outcomes are presented in Table 3.

Coefficients from the fitted interaction model are presented in Table 2, and model-based margins are presented in Figure 1. Corresponding NIHSS-specific adjusted odds ratios and absolute risk differences for successful recanalization, derived from the interaction model, are summarized in Table 4, which reports NIHSS-specific predicted probabilities, odds ratios, and absolute risk differences with corresponding 95% confidence intervals. For completeness, stratum-specific NIHSS marginal effects (per 1-point NIHSS increase) with 95% CIs for each mTICI stratum are presented in eTable 1. The model-based crossover between the mTICI strata occurred at NIHSS = 10.3 (where the recanalization OR = 1) and is reported only as a descriptive feature, not as a significance threshold. Using the Johnson-Neyman procedure (restricted to the observed NIHSS range), statistical evidence of benefit emerged only at the upper end of the observed NIHSS range. In the tabulated NIHSS-specific estimates, the adjusted OR for successful recanalization was 2.34 (95% CI 0.90–6.11) at NIHSS 15 and 3.63 (95% CI 1.07–12.32) at NIHSS 18 (Table 4). To aid clinical interpretation around the crossover, the banded margins analysis for NIHSS 10–12 summarizes the direction and magnitude of the effect in that range; as expected from the figure, point estimates favor recanalization and increase across 10–12, while 95% CIs include 1.0, consistent with limited precision at these severities. Corresponding model-based predicted probabilities at NIHSS 5, 8, 10, 12, 15, and 18

Table 1 Study Cohort Baseline Characteristics and Study Cohort Primary and Secondary Outcomes, Procedural Success, and Safety Events

	Total n = 147	mTICI≥2b n = 85	mTICI≤2a n = 62	*p* Value
Baseline clinical and pre-EVT variables				
Age in years, median (IQR)	77 (65–84)	78 (66–83)	77 (63–85)	0.95
Female sex, n (%)	69 (47)	34 (40%)	35 (56%)	0.05
Baseline NIHSS, median (IQR)	11 (6–17)	10 (6–18)	14 (8–19)	0.04
Baseline ASPECTS, median (IQR)	8 (7–10)	8 (7–10)	8 (7–9)	0.11
Time from onset or last seen well to imaging in minutes, median (IQR)	98 (70–155)	95 (69-155)	107 (73-154)	0.04
Intravenous thrombolysis, n (%)	84 (57)	55 (65%)	29 (47%)	0.03
Outcomes				
Primary outcome				
mRS 0–2 at day 90, n (%)	72 (49)	48 (57)	24 (39)	0.03
Secondary outcomes				
mRS at day 90, median (IQR)	3 (1–5)	2 (1–4)	4 (2–5)	0.02
mRS 0–1 at day 90, n (%)	48 (33)	35 (41)	13 (21)	0.01
PSV at 24 h, median (IQR)	46 (8–82)	59 (19–91)	22 (0–67)	<0.01
Additional outcome and procedural parameters				
mRS 5–6, n (%)	56 (38)	25 (29)	31 (50)	0.01
NIHSS at 24 hours, median (IQR)	8 (2–14)	6 (2–12)	10 (4–17)	<0.01
Safety events sICH, n (%)	10 (6.8)	7 (8.2)	3 (4.8)	0.42

Abbreviations: ASPECTS = Alberta Stroke Program Early CT Score; EVT = endovascular treatment; IQR = interquartile range; mRS = modified Rankin Scale; mTICI = modified Thrombolysis in Cerebral Infarction; NIHSS = National Institutes of Health Stroke Scale; PSV = penumbra salvage volume (baseline penumbra volume minus final infarct volume on follow-up imaging); sICH = symptomatic intracerebral hemorrhage.

are derived from the same interaction model; the associated absolute risk differences between mTICI ≥2b and ≤2a are described in the text to illustrate the magnitude and direction of the effect across the severity spectrum.

Finally, IPW analysis was applied to compare the ATE of achieving successful recanalization (mTICI ≥2b vs ≤ 2a) on functional independence at day 90 within the EVT cohort. As an exploratory, severity-informed summary guided by the

Table 2 Univariable and Multivariable Logistic Regression Analysis With and Without Interaction Term for Primary Outcome Modified Rankin Scale Scores 0–2 at Day 90

	Univariable analysis, outcome mRS at 90 d			Multivariable analysis, outcome mRS at 90 d without interaction term			Multivariable analysis, outcome mRS at 90 d with interaction term		
	OR	95% CI	*p* Value	aOR	95% CI	*p* Value	aOR	95% CI	*p* Value
Primary outcome mRS at 90 d									
NIHSS	0.87	0.83–0.93	<0.01	0.92	0.86–0.99	0.03	0.83	0.73–0.94	<0.01
mTICI≥2b	2.05	1.05–4.00	0.03	1.18	0.47–2.99	0.72	0.14	0.02–1.26	0.08
Interaction term NIHSS x mTICI≥2b	—	—	—	—	—	—	1.19	1.01–1.39	0.03

Abbreviations: aOR = adjusted odds ratio; ASPECTS = Alberta Stroke Program Early CT Score; mRS = modified Rankin Scale; mTICI = modified thrombolysis in cerebral infarction; NIHSS = National Institute of Health Stroke Scale; OR = odds ratio.
Multivariable logistic regression models were adjusted for the following covariates: age, sex, ASPECTS, time from onset to imaging, application of intravenous alteplase, baseline ischemic core volume, and center.
In the interaction model, NIHSS represents the effect of a one-point NIHSS increase among patients with mTICI ≤2a (reference group). The interaction term NIHSS×mTICI≥2b represents the multiplicative change in the NIHSS effect for patients with mTICI ≥2b. Thus, the NIHSS slope in the mTICI ≥2b stratum is obtained by combining the main NIHSS term with the interaction term on the log-odds scale.

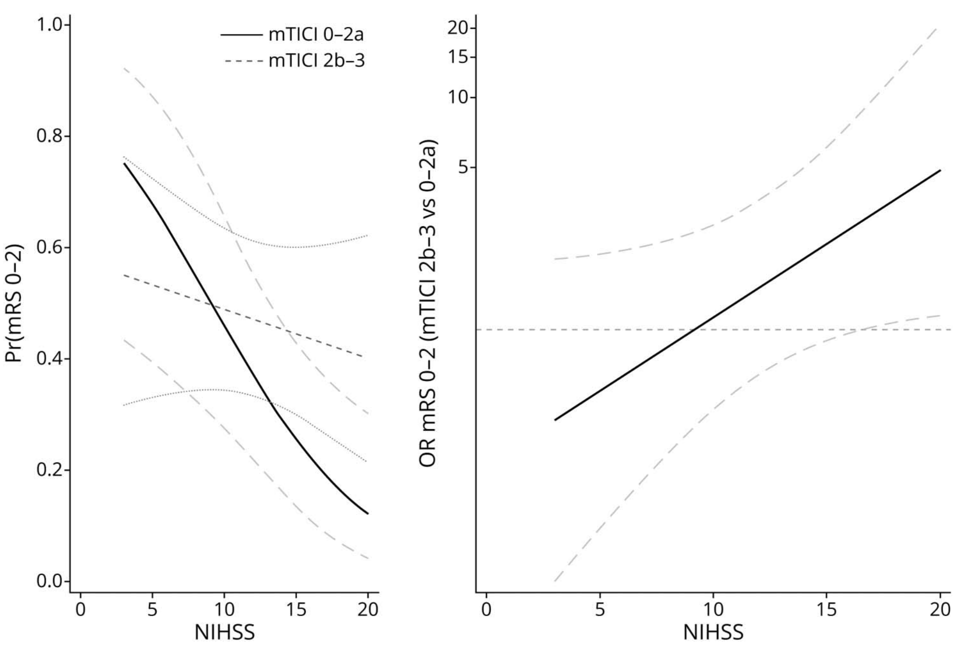
Table 3 Univariable and Multivariable Regression Analyses With and Without Interaction Term for Secondary Outcomes: Penumbra Salvage Volume, and NIHSS Change From Baseline to Follow-Up

	Univariable analysis			Multivariable analysis without interaction term			Multivariable analysis with interaction term		
	ß	95% CI	*p* Value	ß	95% CI	*p* Value	ß	95% CI	*p* Value
PSV									
mTICI≥2b	33.4	6.9 to 59.8	0.01	38.4	11.7 to 65.1	<0.01	101.3	51.6 to 151.1	<0.01
Time onset-imaging	2.1	−1.9 to 6.0	0.29	3.05	−0.9 to 6.9	0.12	19.8	12.0 to 27.5	<0.01
Time onset–imaging in mTICI ≤2a	—	—	—	—	—	—	19.8	12.0 to 27.5	<0.01
Time onset–imaging in mTICI ≥2b (β_time + β_time×mTICI≥2b)	—	—	—	—	—	—	−0.6	−13.6 to 12.4	0.93
Time onset–imaging × mTICI ≥2b (interaction term)	—	—	—	—	—	—	−20.4	−30.9 to −10.0	<0.01
NIHSS Change									
mTICI≥2b	1.2	0.9 to 3.4	0.27	2.2	0.2 to 4.2	0.03	—	—	—
Baseline NIHSS	0.3	0.2 to 0.5	<0.01	0.5	0.3 to 0.6	<0.01	—	—	—

Abbreviations: β = regression coefficient; ASPECTS = Alberta Stroke Program Early CT Score; mTICI = modified Thrombolysis in Cerebral Infarction; NIHSS = National Institutes of Health Stroke Scale; PSV = penumbra salvage volume.
Stratum-specific time effects are reported only for the interaction model; without interaction, the time effect is assumed identical across recanalization strata.
Multivariable linear regression models were adjusted for the following covariates: age, sex, ASPECTS, time from onset to imaging, application of intravenous alteplase, baseline ischemic core volume, and center.

pattern around NIHSS ≈10, ATEs were reported separately for patients with NIHSS >10 and NIHSS ≤10; these categories were chosen for descriptive purposes and do not represent a formal J–N–based threshold. In patients with NIHSS >10, successful recanalization was associated with a higher probability of functional independence (ATE 21.7 percentage points, 95% CI 2.4–40.9, $p = 0.03$) when using the same pretreatment covariates as in the multivariable model. In

Figure 1 Relationship Between Baseline NIHSS, Recanalization Status, and 90-Day Functional Independence



Relationship between baseline NIHSS, recanalization status, and 90-day functional independence. Left: model-based predicted probability of functional independence (mRS 0–2) at 90 days for patients with unsuccessful recanalization (solid line, mTICI 0–2a) vs successful recanalization (dashed line, mTICI 2b–3) across the observed NIHSS range (shown up to NIHSS 20); thin lines denote 95% confidence limits for each curve. Right: corresponding odds ratio for mRS 0–2 comparing mTICI 2b–3 vs 0–2a as a function of NIHSS; thin lines denote 95% confidence limits and the horizontal dashed line marks OR = 1. The right panel is displayed on a logarithmic y-axis. Estimates are derived from the prespecified multivariable logistic regression interaction model with linear NIHSS and an NIHSS×recanalization interaction, adjusted for age, sex, ASPECTS, intravenous thrombolysis, onset/last-known-well-to-imaging time, baseline ischemic core volume, and center (see Statistical Analysis). ASPECTS = Alberta Stroke Program Early CT Score; mRS = modified Rankin Scale; mTICI = modified Thrombolysis in Cerebral Infarction; NIHSS = National Institutes of Health Stroke Scale; OR = odds ratio.

Table 4 NIHSS-Specific Predicted Functional Independence by Recanalization Status With 95% CI

NIHSS	Pr(mRS 0–2), mTICI 0–2a (95% CI)	Pr(mRS 0–2), mTICI 2b–3 (95% CI)	aOR mRS 0–2 (2b–3 vs 0–2a) (95% CI)	ARDs (2b–3 minus 0–2a) (95% CI)
5	0.678 (0.421–0.935)	0.533 (0.325–0.741)	0.54 (0.14–2.12)	−0.145 (−0.378 to 0.088)
8	0.549 (0.321–0.778)	0.506 (0.339–0.674)	0.84 (0.29–2.42)	−0.043 (−0.294 to 0.208)
10	0.459 (0.259–0.658)	0.489 (0.339–0.638)	1.13 (0.45–2.82)	0.030 (−0.207 to 0.267)
12	0.371 (0.198–0.544)	0.471 (0.328–0.614)	1.51 (0.64–3.57)	0.100 (−0.115 to 0.315)
15	0.255 (0.105–0.404)	0.444 (0.289–0.600)	2.34 (0.90–6.11)	0.189 (−0.069 to 0.447)
18	0.165 (0.031–0.300)	0.418 (0.230–0.607)	3.63 (1.07–12.32)	0.253 (0.014 to 0.492)

Abbreviations: aOR = adjusted odds ratio; ASPECTS = Alberta Stroke Program Early CT Score; mTICI = modified thrombolysis in cerebral infarction; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale.
Values are derived from the primary multivariable logistic regression interaction model with mRS 0–2 at 90 days as the outcome, baseline NIHSS modeled linearly, and an NIHSS×recanalization interaction term, adjusted for prespecified covariates (age, sex, ASPECTS, IV thrombolysis, onset/last-known-well-to-imaging time, baseline ischemic core volume, and center). Predicted probabilities of 90-day functional independence (mRS 0–2) are shown for each recanalization stratum (mTICI 0–2a and mTICI 2b–3) at representative NIHSS values. Odds ratios compare mTICI 2b–3 vs 0–2a at the specified NIHSS value. Absolute risk differences (ARDs) represent the difference in predicted probability of mRS 0–2 between mTICI 2b–3 and mTICI 0–2a. Confidence intervals for probabilities and absolute risk differences were derived on the logit scale (delta method) and transformed to the probability scale; confidence intervals for odds ratios were computed on the log-odds scale and exponentiated.

contrast, for patients with NIHSS ≤10, successful recanalization was not associated with a clear functional benefit (ATE 1.2 percentage points, 95% CI −22.0 to 24.0, p = 0.91). These weighted estimates were concordant in direction and magnitude with the covariate-adjusted interaction model and its margins. The distribution of functional outcome at 90 days by recanalization status and NIHSS strata is presented in Figure 2. The results of the supplementary 3-category TICI sensitivity analysis are presented in eTable 2 and were directionally consistent with the primary binary recanalization model.

Secondary End Points

Penumbra Salvage Volume

In univariable linear regression analysis, only successful recanalization was associated with PSV (ß: 33 mL, 95% CI 6.9–59.8, p = 0.01). In addition, potential interaction terms between vessel recanalization status and all available baseline parameters were tested. Only time from symptom onset to imaging showed a significant interaction with recanalization (Table 3). Because the PSV model included a time-from-onset-to-imaging × recanalization interaction, the reported time coefficient reflects the association in the mTICI ≤2a reference group; the corresponding time effect in mTICI ≥2b is presented in Table 3 as the linear combination β_time + β_time×mTICI ≥2b with 95% confidence intervals. Plotting the relationship of time from symptom onset to imaging and recanalization status, recanalization was consistently associated with penumbra salvage and constant PSV over time (see red graph in Figure 3—constant PSV for time from onset to imaging shown on the x axis); however, compared with patients not achieving recanalization, the statistical difference was only observed up to 2.5 hours from onset (no CI overlap up to 2.5 hours from onset) (Figure 3).

Second, the effect of recanalization on PSV was compared in patients with NIHSS >10 vs patients with NIHSS ≤ 10 based on IPW. In patients with NIHSS >10, recanalization was associated with significant penumbra salvage (PSV 45 mL, p = 0.03). For patients with NIHSS ≤ 10, recanalization was not associated with significant penumbra salvage (PSV 23.5 mL, p = 0.2).

NIHSS Change Within 24 hours

In univariable linear regression analysis, only admission NIHSS was associated with NIHSS change from admission to 24 hours (Table 3). In multivariable linear regression analysis without interaction, successful recanalization was associated with a 2.2-point greater NIHSS decline by 24 hours (β 2.2, 95% CI 0.2–4.2, p = 0.03). Admission NIHSS was also independently associated with greater 24-hour NIHSS decline (β 0.46 per point, 95% CI 0.31–0.61, p < 0.001).

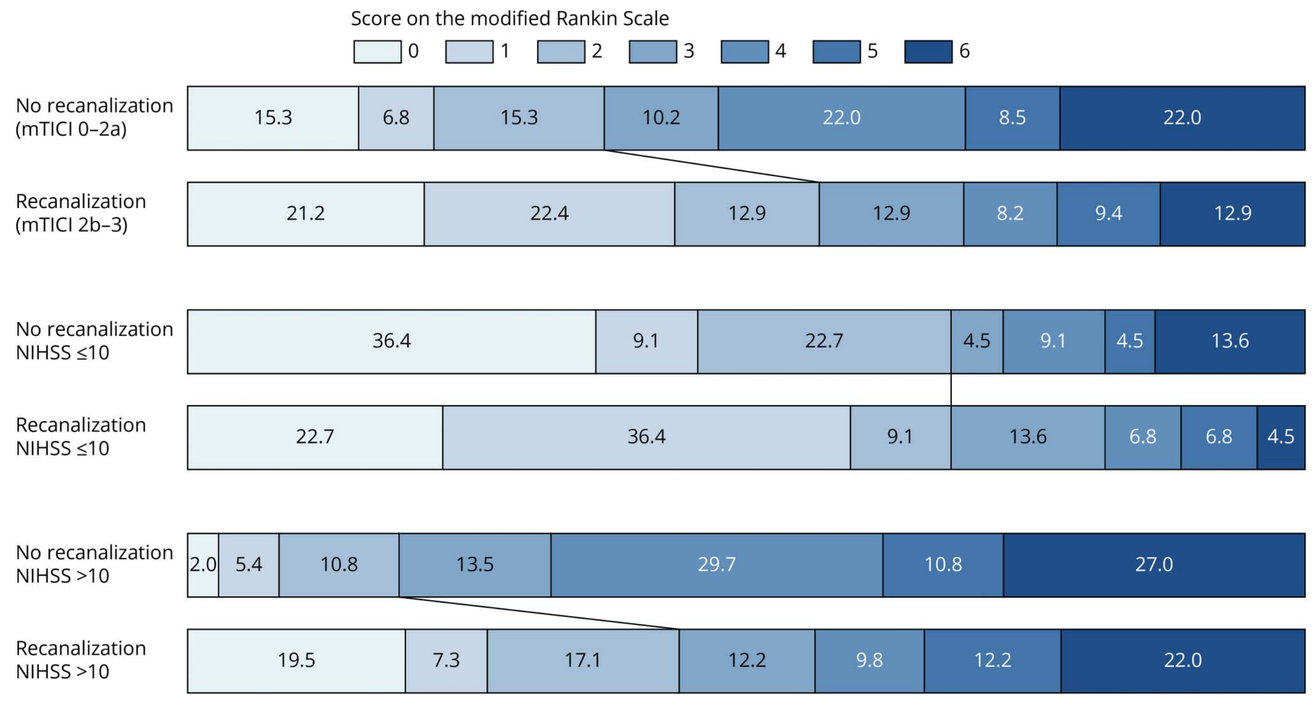
Symptomatic Intracerebral Hemorrhage

The rate of sICH was 8.2% in successfully recanalized patients compared with 4.8% in patients whom recanalization could not be achieved (p = 0.42). In univariable and multivariable logistic regression analysis, no association between sICH and any baseline variable was identified.

Discussion

In this multicenter M2-only EVT cohort, we investigated whether the association between successful recanalization and functional outcome is modified by baseline clinical or imaging variables. The main finding was that successful EVT was associated with improved functional outcomes and greater penumbra salvage volume, and that this association was strongly dependent on baseline stroke severity: NIHSS was the only variable that showed a significant interaction with recanalization status. Patients with higher NIHSS had markedly worse outcomes when recanalization failed, whereas this adverse severity gradient was attenuated when mTICI ≥2b

Figure 2 Distribution of Functional Outcome at 90 Days

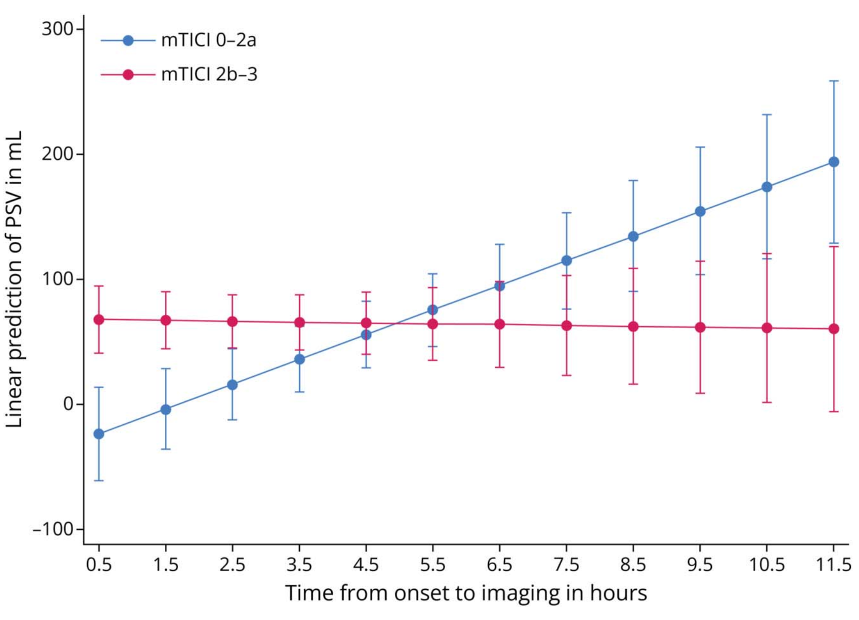


Bar diagrams to show the distribution of final functional outcome at day 90 based on the modified Rankin Scale score separately for patients with vs without vessel recanalization, and NIHSS subgroup (>10 vs ≤ 10), based on the model-based crossover region. mTICI = modified thrombolysis in cerebral infarction; NIHSS = National Institutes of Health Stroke Scale.

was achieved. By contrast, age and ASPECTS, although independently associated with favorable outcome, did not demonstrate treatment effect modification. Intravenous alteplase was neither significantly associated with better functional outcome nor modified the effect of EVT.

These findings are consistent with previous observational M2 studies suggesting that the magnitude of benefit associated with successful reperfusion/recanalization is larger in patients with more severe baseline deficits. In our cohort, the fitted curves crossed around NIHSS ≈10 as a descriptive transition

Figure 3 Relationship of Time From Onset to Imaging and Recanalization Status With Penumbra Salvage Volume (PSV)



Multivariable linear regression analysis including the interaction term between time from onset to imaging and recanalization status showing the association to PSV to follow-up. mTICI = modified thrombolysis in cerebral infarction.

zone, whereas formal statistical evidence of benefit emerged only at the upper end of the observed NIHSS range. This pattern should not be interpreted as a fixed threshold, but it supports the concept of severity-informed rather than uniform patient selection in M2 occlusion.[9-11]

The interaction between onset-to-imaging time and recanalization influenced PSV but not functional independence, indicating that imaging-based selection alone may have limited discriminatory power if clinical severity is ignored. Overall, these results suggest that NIHSS should be considered alongside imaging and other clinical factors when evaluating EVT candidacy in M2 occlusions, rather than interpreted as a standalone or prescriptive selection rule.[12] Future and ongoing M2/DMVO studies that incorporate combined clinical severity and imaging-based selection criteria will be important to prospectively test whether severity-informed triage can identify subgroups with net functional benefit while minimizing hemorrhagic and procedural risk.[13,14] While recent trials have questioned routine EVT in DMVO, our data support a selective approach in which patients with more significant neurologic deficits may still benefit from timely recanalization, consistent with other observational work.[9,15-17]

Importantly, EVT was not associated with a statistically significant increase in sICH in this cohort, although previous studies have reported a higher likelihood of vessel perforation in DMVO procedures, underscoring the need to carefully weigh individual risks and benefits.[18]

The interaction between NIHSS and recanalization is summarized in Table 2 and illustrated in Figure 1. In the logistic model, the NIHSS×recanalization term was significant, consistent with a steeper severity-outcome gradient when recanalization fails (mTICI ≤2a) and a flattened gradient when recanalization succeeds (mTICI ≥2b). Model-based margins showed that the predicted probabilities for mRS 0–2 with and without successful recanalization were similar at lower NIHSS, that the 2 curves crossed around NIHSS ≈10, and that the estimated contrast between recanalization strata became larger at higher NIHSS. The margins plot displays probabilities, not odds; because baseline probability falls with NIHSS and the probability scale is bounded, a growing odds ratio can correspond to only modest visual curve separation. Inverse probability–weighted analyses, which estimated the average treatment effect of successful recanalization under the same set of pretreatment covariates showed a concordant pattern, with no clear benefit at lower NIHSS values and a higher probability of independence with recanalization at higher NIHSS values. This consistency across modeling approaches supports a severity-dependent pattern, while still acknowledging limited precision in the lower NIHSS range. eTable 1 complements this presentation by quantifying, within each recanalization stratum, the model-based association of a 1-point NIHSS increase with functional independence at representative NIHSS values. eTable 1 presents exploratory spline-based local NIHSS slopes within each recanalization stratum. These supplementary estimates should be interpreted only as sensitivity analyses of possible nonlinearity; the primary manuscript inferences are based on the prespecified linear NIHSS×recanalization interaction model.

The main challenge in guiding treatment decision-making in DMVO stroke is the heterogeneity of recent trial populations, including broad NIHSS criteria, wide time windows, mixed vessel territories, and limited imaging-based enrichment. Penumbral volume may be one relevant discriminator in M2 occlusion, as EVT benefit has been linked to larger tissue-at-risk profiles that likely reflect the variable territory supplied by the affected M2 branch.[13] CTP-based penumbral imaging may therefore support imaging triage,[19-21] but other NECT-derived, perfusion-derived, and CTA-derived markers of tissue injury, edema progression, and collateral status remain insufficiently validated as treatment-effect modifiers in DMVO and require further study.[22-30]

It is notable that across the major recent trials, the rate of sICH was consistently higher in patients treated with thrombectomy compared with those managed with best medical treatment alone.[31] In the ESCAPE-MeVO trial, sICH was strongly associated with mortality at 90 days, raising concerns about procedural safety in this subgroup. The trial authors suggested that alternative endovascular strategies might offer better outcomes, particularly as stent retrievers were mandated as the first-line device in ESCAPE-MeVO and used in the vast majority of cases in the DISTAL trial. In addition, both trials reported relatively modest reperfusion rates—approximately 75% in ESCAPE-MeVO and 72% in DISTAL—which fall short of those achieved in earlier large-vessel occlusion trials.[32] It remains uncertain whether these lower reperfusion rates reflect technical constraints inherent to current device use in distal occlusions,[33,34] or whether treatment decisions—potentially influenced by patient age or milder stroke severity—contributed to the observed outcomes.

While this study provides important insights into severity-based treatment stratification, it did not include harmonized CTA-based anatomical subclassification of M2 occlusions, such as proximal vs distal segments or branch dominance/codominance. These features may influence procedural complexity and outcome, but their definitions and assessment vary across raters and centers, limiting their reliability and generalizability as retrospective multicenter adjustment variables.[35,36]

The primary end point of this study was functional independence defined as mRS 0–2 at day 90 in contrast to previous DMVO trials that used mRS 0–1 as primary end point. This cohort included only M2 occlusions (no M3/M4 or other DMVO) and showed a comparatively high median NIHSS of 11, underscoring the substantial deficit burden in this group. In recent distal/medium-vessel occlusion trials, functional independence has been generally modest despite more distal occlusion sites: in DISTAL, approximately 55%–

57% of patients achieved mRS 0–2 in both EVT and control arms, with about 15%–17% having poor outcome (mRS 5–6), and in ESCAPE-MeVO only 54.1% of EVT patients reached mRS 0–2[1,3]. By contrast, in the HERMES meta-analysis of proximal LVO trials, mRS 0–2 occurred in ≈46% of EVT patients vs 26.5% of controls, with poor outcome (mRS 5–6) in ≈22% and 32%, respectively. These figures highlight that—even in more distal occlusions—there remains a substantial risk of disability and death despite treatment. As a secondary imaging end point, we investigated PSV as the difference between baseline tissue-at-risk (hypoperfusion minus core) and final infarct volume on follow-up CT.[7,37] Recanalization was consistently associated with stable PSV over time, whereas PSV declined in patients without recanalization; however, the statistical effect was only evident up to approximately 2.5 hours from onset, suggesting a time-critical relationship between penumbra salvage and recanalization.[38] The time×recanalization interaction for PSV should be interpreted cautiously. Because onset-to-imaging time is not equivalent to time-to-recanalization and treatment selection may depend on imaging profiles, the stratum-specific time associations likely reflect case-mix and selection effects rather than a causal time effect. Importantly, the main PSV finding was robust: successful recanalization was consistently associated with substantially greater penumbra salvage. Considering the broad time windows of the DMVO trials (up to 24 hours in DISTAL and 12 hours in ESCAPE-MeVO), time from onset may be another relevant factor for treatment decision-making, generally encouraging EVT in hyperacute cases.

This study has several limitations. First, the retrospective, observational design introduces the possibility of selection and confounding biases, and our analyses remain associational; they cannot establish causal effects, even with multivariable adjustment and IPW. IPW estimates an average treatment effect under assumptions of no unmeasured confounding, adequate overlap, and correct propensity model specification, and violations of these assumptions cannot be excluded. Second, prestroke functional status (mRS) and detailed vascular risk factors were not available in a harmonized manner across both centers, and residual confounding by baseline disability and comorbidity cannot be excluded. In addition, even when recorded, prestroke functional status may be difficult to ascertain reliably in the acute stroke triage setting, which further limits its utility for retrospective adjustment. Third, although the M2-only design reduces anatomical heterogeneity compared with prior DMVO trials, the sample size—particularly in lower NIHSS strata—is modest for interaction analyses and contributes to wide confidence intervals, limiting the precision of severity-specific estimates. Fourth, imaging and treatment practices were not fully standardized across centers, despite adjustment for site, and our findings may not generalize to more distal occlusions or to other health-care settings. Fifth, detailed CTA-based anatomical subclassification of M2 occlusions (proximal vs distal segment and branch dominance/codominance) was not available in a harmonized manner across centers and could not be incorporated into the primary models. Moreover, even when assessed clinically, these anatomical labels are not standardized and may vary by operator and local terminology (including differing segment definitions and criteria for "dominant"/"codominant"/"nondominant" branches), which limits their reliability as a uniform selection or adjustment variable in retrospective multicenter EVT cohorts and may contribute to residual confounding. Similarly, procedural variables such as device strategy, number of passes, and puncture-to-recanalization time were not available in a harmonized manner and could not be incorporated; residual confounding by unmeasured procedural factors therefore remains possible. Finally, our analyses were not designed to identify an exact NIHSS cutoff for clinical decision-making; the intention was to characterize how clinical severity modifies the association between recanalization and outcome. Continuous interaction analysis showed a larger estimated effect size (contrast between recanalization strata) at higher NIHSS, while Johnson-Neyman probing indicated that formal statistical evidence of benefit emerges only at higher NIHSS values and depends partly on sample size and variance. Sensitivity analyses, including IPW, were directionally concordant, but do not convert this pattern into a rigid treatment rule. Thus, any NIHSS value (including ≈10) should be regarded as a pragmatic heuristic summarizing the observed pattern, rather than a fixed threshold, and our data are best interpreted as supporting the broader principle that patients with more severe deficits are most likely to benefit from EVT in M2 occlusion.

In conclusion, the benefit of EVT for acute M2 occlusions is not uniform but appears to be strongly influenced by initial clinical severity. In our M2 cohort, the magnitude of the association between successful recanalization and functional independence was larger at higher NIHSS, while clear benefit was not demonstrable in milder strokes. NIHSS therefore emerges as a robust marker of treatment effect modification and should be incorporated—together with imaging and procedural considerations—into future EVT selection strategies for DMVO. These observational findings support a severity-informed, individualized approach to EVT rather than uniform eligibility criteria and underscore the need for prospective studies to confirm severity-based selection in M2 occlusions.

Author Contributions

G. Broocks: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. H.C. Kniep: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. M.G. Lansberg: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. C. Heitkamp: drafting/revision of the manuscript for content, including medical writing for content. V. Yedavalli: drafting/revision of the manuscript for content, including medical writing for content. H. Krähling: drafting/revision of the manuscript for content, including medical writing for content. E. Hoffmann: major role in the acquisition of data. L. Meyer: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. A. Kemmling: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. P. Stracke: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. G.W. Albers: major role in the acquisition of data; analysis or interpretation of data. J. Fiehler: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. J. Minnerup: study concept or design; analysis or interpretation of data. K. Alfke: analysis or interpretation of data. J.J. Heit: major role in the acquisition of data; analysis or interpretation of data. T.D. Faizy: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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